KATO KSP40 4” x 4” Centrifugal Pump coupled to KATO 186F Diesel Engine

Test Report No.	:	2014 - 26
Date of Test	:	March 14, 2014
Test Valid Until	:	March 14, 2019
Brand	:	KATO
Model	:	KSP40 4” x 4”
Serial No.	:	No Data
Type and Size	:	Single-end suction, 100 x 100 mm, Self-Priming Centrifugal pumpset
Manufacturer	:	KATO INTERNATIONAL, INC. Valenzuela City
Test Requested by	:	TRAMAT MERCANTILE, INC. 747-749 Sabino Padilla, St., Sta. Cruz, Manila
Machine Classification	:	Commercial

SUMMARY

The KATO KSP40 4”x4” Self-priming Centrifugal pump coupled to KATO 186F Diesel Engine was tested for performance and suction condition. It was tested at the recommended full throttle speed setting of the engine. The engine and pump both usd double groove B-section pulleys with 102 mm (4.0 inches) and to pieces of V-belts. The engine and pump were mounted on a steel base.

The Maximum Total Head (H) was 28.2 m at zero discharge condition. The shaft speed of the pump and engine at this point were 2111 and 3521 rpm, respectively, with Engine Fuel Consumption Rate of 3.10 L/h. The Maximum Discharge Capacity of 22.4 L/s occurred at 7.80 m Total Head with Fuel Consumption Rate of 3.08 L/h. The shaft speed of the pump and engine at this point were 1684 and 2708 rpm, respectively. The Maximum System Efficiency of 7.22% occurred at 17.4 L/s Discharge Capacity.

The suction condition of the pump was also tested. With zero discharge, the Maximum Total Suction Lift was 6.21 m. The Net Positive Suction Head Available at this point was 3.71 m.

PURPOSE AND SCOPE OF TEST

The purpose of the test was to establish the pump performance which included the Fuel Consumption Rate and the Total Head at varying discharge capacities and suction conditions. The pump was mounted on the laboratory test set-up shown in **Figure 1.**

The test was conducted on March 14, 2014 at the AMTEC Test Laboratory, UPLB, College, Laguna.

Figure 1. KATO KSP40 4”x4” on the test set-up

METHOD OF TEST

The pump was tested following the “Philippine Agricultural Engineering Standards, PAES 115:2000, Agricultural Machinery – Centrifugal, Mixed-Flow, Axial-Flow Water Pumps – Methods of Test”.

DESCRIPTION OF THE PUMP

The KATO KSP40 Self-priming Centrifugal Pump shown in **Figure 2**, was a single-end suction, 100 mm x 100 mm, centrifugal pump close-coupled to a 7.46 kW (10 Hp) KATO 186F diesel engine. It was mounted on a steel base. The suction inlet and discharge outlet were threaded and designed for a 100 mm diameter pipe. The impeller was a semi-open type made of brass material connected to the pump shaft through a thread. It was housed in a cast iron dual volute casing enclosed in an aluminum chamber. The priming inlet was located on top of the pump beside the discharge outlet. The suction inlet was provided with a swing type one-way valve to retain the sufficient amount of liquid inside the chamber necessary to prime the pump. Located on top of the pump was the discharge outlet. The stuffing box used a mechanical seal.

The primemover, a 7.46 kW KATO 186F diesel engine was a four-stroke, air-cooled, single vertical cylinder diesel engine. Fuel feed system was of gravity type with the fuel tank located on top of the engine. It utilized a dual type air cleaner with pleated paper and foam element. Lubrication system was force-feed lubricating type. Cooling was done by air forced by a fan/flywheel encased in a metal shroud. It was started by a rope-recoil system. A muffler was used in the exhaust system. **Table 1** shows the detailed specifications of the pump.

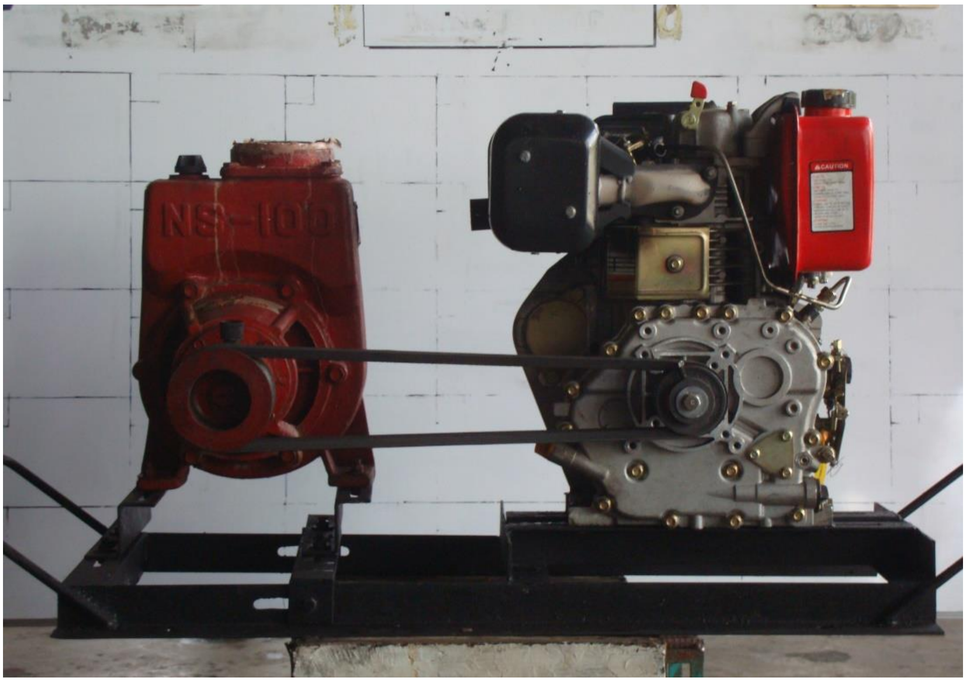

Figure 2. KATO KSP40 4” x 4” Self-priming Centrifugal Pump coupled to KATO 186F Diesel Engine

PUMPSET SPECIFICATIONS

Table 1. Detailed specifications of the KATO KSP40 4” x 4” Centrifugal Pump coupled to KATO 186F Diesel Engine

1.	Brand		:	KATO
2.	Model		:	KSP40 4” x 4”
3.	Serial No.		:	No Data
4.	Make		:	KATO INTERNATIONAL, INC. Valenzuela City
5.	Type		:	Single-end suction, Self-priming centrifugal pump
6.	Size (mm)		:	100 x 100
7.	Overall Dimension and Weight (Including mounting frame and engine)			
	7.1	Length (mm)	:	1220
	7.2	Width (mm)	:	480
	7.3	Height (mm)	:	595
	7.4	Weight (kg)	:	102
8.	Impeller			
	8.1	Type	:	Semi-open
	8.2	Diameter (mm)	:	190
	8.3	Width (mm)	:	29
	8.4	No. of vanes	:	5
	8.5	Material	:	Brass
9.	Suction Inlet			
	9.1	Type	:	Threaded flange
	9.2	Diameter (mm)	:	100
	9.3	Material	:	Aluminum
10.	Discharge Outlet			
	10.1	Type	:	Threaded flange
	10.2	Diameter (mm)	:	100
	10.3	Material	:	Aluminum

Table 1. (Cont’n)

11.	Casing		
	11.1 Type	:	Double volute
	11.2 Material	:	Cast Iron
12.	Stuffing box	:	Mechanical Seal
13.	Rotation	:	Counter clockwise when facing the suction inlet
14.	Rated speed (rpm)	:	2400
15.	Rated Maximum Discharge Capacity (L/s)	:	No Data
16.	Rated Maximum Total Head (m)	:	No Data
17.	Primemover (based on Plate Rating)		
	17.1 Brand	:	KATO
	17.2 Model	:	186F
	17.3 Serial No.	:	5100918 18207
	17.4 Make	:	No Data
	17.5 Type	:	Four stroke cycle, air-cooled, single vertical cylinder diesel engine
	17.6 Rated Output Power (kW)		
	Maximum	:	7.46 @ 3600 rpm
	Continuous	:	No Data
	17.8 Bore x Stroke (mm)	:	No Data
	17.9 Displacement Volume (cc)	:	418
	17.10 Cooling System	:	Air-cooled
	17.11 Lubricating System	:	Force-feed
	17.12 Starting System	:	Rope-recoil
	17.13 Air Cleaner	:	Dual type (Foam and pleated paper element)
	17.14 Exhaust System	:	Muffler

PERFORMANCE TEST

Results of performance test are shown in **Table 2** and plotted in **Figure 3**.

Pumpset Tested	:	KATO KSP40 4” x 4”
Date of Test	:	March 14, 2014
Test Condition		
Ambient Temperature (^oC)	:	31.9
Atmospheric Pressure (mb)	:	1013
Pumping Water Temperature (^oC)	:	29.0

Table 2. Results of Performance Test

ENGINE SHAFT SPEED (rpm)	PUMP SHAFT SPEED (rpm)	FUEL CONSUMP-TION (L/h)	TOTAL HEAD (m)	DISCHARGE CAPACITY (L/s)	WATER POWER (kW)	INPUT POWER (kW)	SYSTEM EFFICIENCY (%)
3521	2111	3.10	28.2	0	0	32.3	0
3398	2063	3.50	25.4	5.18	1.29	36.4	3.54
3288	2001	3.64	23.2	8.66	1.97	37.9	5.21
3164	1953	3.51	20.7	11.8	2.40	36.5	6.57
3092	1919	3.45	18.8	13.6	2.50	35.9	6.98
3016	1864	3.39	16.4	15.8	2.54	35.2	7.21
2915	1813	3.29	14.5	17.4	2.47	34.2	7.22
2874	1783	3.23	12.8	18.6	2.35	33.6	6.98
2802	1747	3.16	11.0	21.5	2.31	32.9	7.03
2749	1707	3.11	9.16	22.3	2.00	32.3	6.20
2708	1684	3.08	7.80	22.4	1.71	32.1	5.33

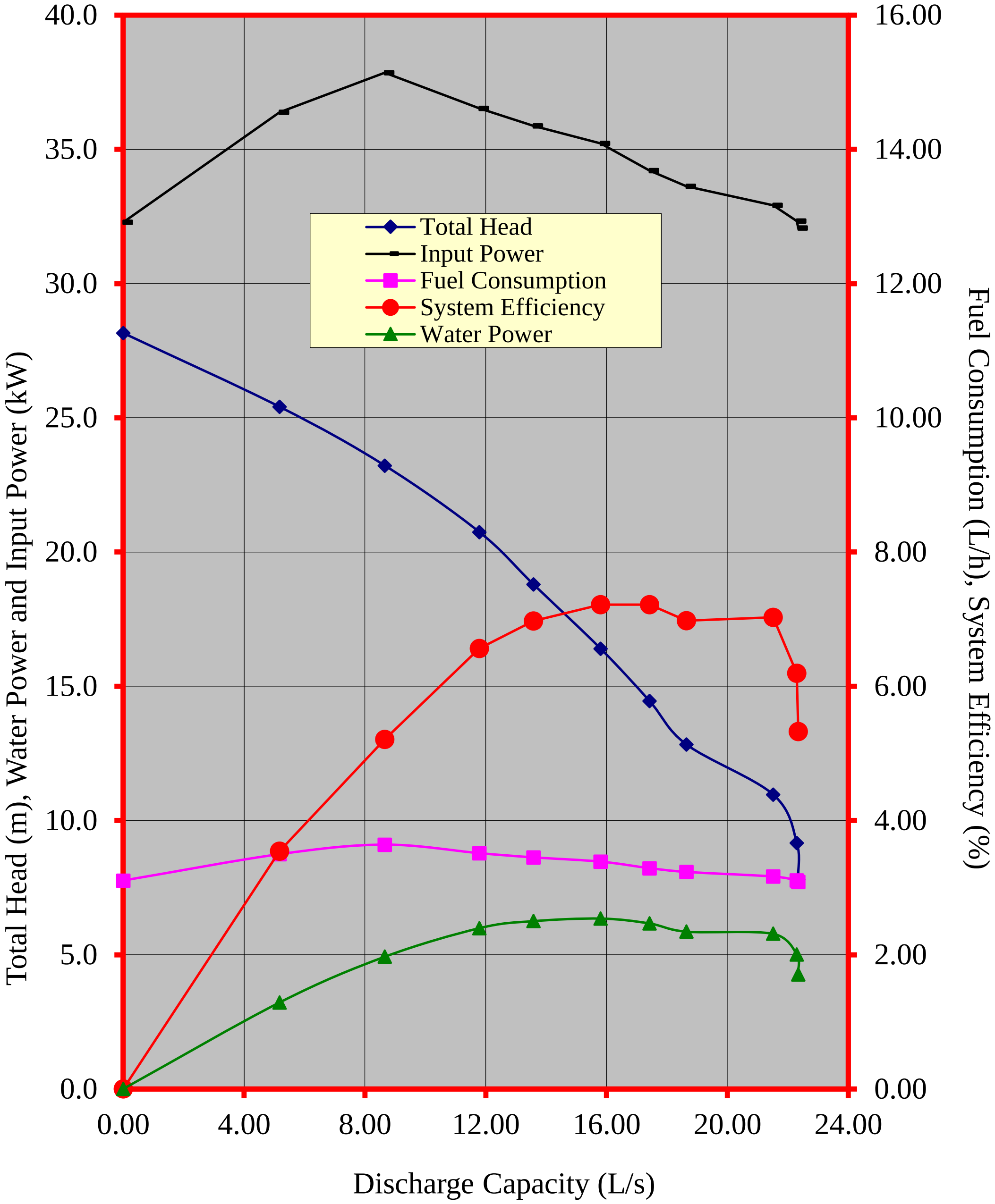

Figure 3. Performance characteristic curves of KATO KSP40 4"X4" self-priming centrifugal pump

CAVITATION TEST

Results of Cavitation Test are shown in **Table 3**.

Pumpset Tested : KATO KSP40 4” x 4”

Date of Test : March 14, 2014

Test Condition

Ambient Temperature (^{o}C) : 31.3
Atmospheric Pressure (mb) : 1013
Pumping Water Temperature (^{o}C) : 29.0

Table 3. Results of Cavitation Test

ENGINE SHAFT SPEED (rpm)	PUMP SHAFT SPEED (rpm)	TOTAL SUCTION LIFT (m)	NET POSITIVE SUCTION HEAD AVAILABLE (m)	TOTAL HEAD (m)	DISCHARGE CAPACITY (L/s)
2550	1653	1.00	8.92	7.75	22.3
2546	1658	1.79	8.13	8.38	20.8
2562	1679	3.19	6.73	9.63	19.7
2598	1706	4.59	5.33	10.8	18.3
2677	1766	6.21	3.71	11.9	0

DATA COMPUTATION

ANNEX 1. Formula used during calculation and testing

1. Total Head (m), H

$$TDH = H_d - H_s$$

Where: $$H_d = \frac{P_2}{\gamma} + \frac{V_2^{\,2}}{2g} + Z_2$$

$$H_s = \frac{P_1}{\gamma} + \frac{V_1^2}{2g} + Z_1$$

H_d = Total Discharge Head (m)

H_s = Total Suction Lift/Head (m)

$\frac{P_2}{\gamma}$ = Pressure gage reading on the discharge pipe at gage connection converted to meter

$\frac{P_1}{\gamma}$ = Vacuum gage reading on the suction pipe at the point of gage connection converted to meter

$\frac{V_2^2}{2g}$ = Velocity head on the discharge pipe converted to meter. Computed by determining the velocity of liquid on the pipe by formula, V = Q/A where V, velocity of liquid inside the pipe in meter per second, Q, Discharge rate of the pump in liters per second, and A, Cross-sectional area of the pipe in square meter.

$\frac{V_1^2}{2g}$ = Velocity head on the suction pipe converted to meter

Annex (Cont’n)

Z_2 = The distance from the center of the pressure gage face to the centerline of the pump in meters

Z_1 = The distance from the center of the vacuum gage face to the centerline of the pump in meters

2. Discharge Capacity, Q, in liters per second (L/s) is determined through gravimetric method using a platform scale and a stopwatch.

3. Water Power, WP, is the output power of the pump in kilowatt (kW) and computed using:

$$WP = \frac{H \, X \, Q}{102}$$

4. Input Power, IP, is the power on the input shaft of the pump. This is the power in kilowatt required to drive the pump and is determined by the formula:

$$IP = \frac{T \, X \, N}{974}$$

Where:

T = Torque in kilogram-meter, on the input shaft of the pump and determined by a dynamometer

N = Input shaft angular speed in RPM

5. Pump Efficiency, η,

$$\eta = \frac{WP}{IP}$$

6. Net Positive Suction Head Available, NPSHA. This is the net force (converted to meter) needed to push the water inside the pipe and computed by the formula:

$$NPSHA = P_{atm} - P_{vp} - H_s$$

Where:

P_{atm} = Atmospheric pressure during the test converted to meters
P_{vp} = Vapor pressure of the test liquid at certain temperature converted to meters

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N.B.

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